

Supplementary File for Ultra-Short-Term Photovoltaic Ramp Forecasting via Physics-Guided Generative Sky Video Prediction

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I. DETAILED HYPERPARAMETERS FOR MODEL TRAINING

A. Hyperparameter Settings for the PhyDNet Module

TABLE I lists the hyperparameters used to train the PhyDNet module.

TABLE I
HYPERPARAMETERS OF PHYDNET

Hyper-parameter	Value / Setting
Batch size	32
Learning rate	10^{-4}
Optimizer	Adam
Number of epochs	300
History frames	16
Forecast frames	16
Frame-loss weight α_F	0.84
Moment-loss weight λ_{Phy}	1.0
Teacher-forcing transition epochs	15 and 30
LR scheduler	ReduceLROnPlateau
Scheduler patience	3 epochs
Scheduler factor	0.3
Nonlinearity	LeakyReLU (0.2)
Normalization	GroupNorm

B. Hyperparameter Settings for the Video-Conditional Diffusion Module

TABLE II lists the hyperparameters used to train the video-conditional diffusion module.

TABLE II
HYPERPARAMETERS OF VIDEO CONDITIONAL DIFFUSION

Hyper-parameter	Value / Setting
Training steps	100000
Batch size	8
Trainer	AdamW
Learning rate	2×10^{-4}
Diffusion timesteps	1000
Beta schedule	Cosine
Base channel dim	32
Input convolution kernel	$7 \times 7 \times 7$
Sinusoidal embedding dim	32
Number of resolutions	4
Conditioning frames	4 observed + 16 coarse
Refined output frames	16
Self-attention by resolution	Temporal, spatial, full, full
Cross-attention locations	All four resolutions and middle block
Mid-block attention heads	4
Cross-attention heads	4
ResNetBlock3D norm	RMSNorm
Activation	SiLU

C. Hyperparameter Settings for the RaPVFormer Module

TABLE III lists the hyperparameters used to train the RaPVFormer module.

TABLE III
HYPERPARAMETERS OF RAPVFORMER

Hyper-parameter	Value / Setting
Image encoder output dim	128
Transformer model dim	256
Transformer heads	4
PV embedding dim	16
Activation functions	SiLU
Self-attention feedforward dim	512
Cross-attention feedforward dim	1024
Transformer dropout	0.1
Optimizer	Adam
Learning rate	10^{-4}
Batch size	64
Number of epochs	30
α_S in slope weighting	1.0
Focal focusing parameter γ	2.0
Ramp-class tolerance	5% of rated capacity
Real/generated view weights	0.5/0.5
ω_P in loss function	1.0
ω_S in loss function	0.2
ω_R in loss function	0.5
LR Scheduler	ReduceLROnPlateau
Scheduler patience	3 epochs
Scheduler factor	0.3

II. DETAILED TRAINING AND INFERENCE PROTOCOL

A. PhyDNet Loss and Autoregressive Decoding

PhyDNet is trained from 16 observed frames to predict the following 16 frames. In addition to the future-frame forecasts, the 15 one-step transitions within the observed sequence are reconstructed, yielding 31 supervised prediction steps per clip. The frame loss at step τ is

$$\mathcal{L}_{\text{frame}}^{(\tau)} = \alpha_F \left[1 - \text{SSIM}(\hat{I}_\tau, I_\tau) \right] + (1 - \alpha_F) \left\| \hat{I}_\tau - I_\tau \right\|_1, \quad (1)$$

where $\alpha_F = 0.84$. Let $F_{:,b}$ denote the bank of learned 7×7 physical convolution kernels associated with input channel b , $\text{K2M}(\cdot)$ the kernel-to-moment transform, and C the prescribed differential-moment basis. The physical regularizer and the total training objective are

$$\mathcal{L}_{\text{moment}} = \sum_b \text{MSE}(\text{K2M}(F_{:,b}), C), \quad (2)$$

$$\mathcal{L}_{\text{PhyDNet}} = \frac{1}{31} \sum_{\tau=1}^{31} \mathcal{L}_{\text{frame}}^{(\tau)} + \lambda_{\text{Phy}} \mathcal{L}_{\text{moment}}, \quad (3)$$

with $\lambda_{\text{Phy}} = 1.0$. The teacher-forcing schedule uses transition epochs 15 and 30: the reverse-scheduled sampling probability for the observed encoder sequence increases toward one after epoch 15, whereas the decoder teacher-forcing probability decreases to zero by epoch 30. During validation, testing, and deployment, the decoder receives only its preceding RGB

prediction and the precomputed solar-position mask for the next time step; no future observed RGB frame is supplied.

B. Video-Conditional Diffusion Attention

The diffusion model is conditioned on the four most recent observed RGB frames and the 16-frame coarse forecast produced by PhyDNet, and outputs a refined 16-frame forecast. The historical and coarse-forecast streams are encoded separately; their features are concatenated and fused before being injected into the denoising backbone by cross-attention. Across the four spatial resolutions, the self-attention modes are temporal, spatial, full, and full, respectively. Cross-attention is enabled at all four resolutions and in the middle block.

The cross-attention operator returns only the attention increment, without an internal residual connection. Consequently, the surrounding block adds the residual exactly once. For example, the gated upsampling fusion is implemented as

$$O_{\text{up}} = O_{\text{self}} + g \odot \text{Attn}_{\text{cross}}(O_{\text{self}}, C), \quad (4)$$

where C is the fused condition feature and g is a learned gate. This convention prevents the gated main branch from being added a second time.

C. RaPVFormer Labels and Dual-View Training

For a 16-step target sequence, the auxiliary ramp label at the first forecast step includes the observed-to-forecast boundary,

$$\begin{aligned} \Delta P_1 &= P_{t+1} - P_t, \\ \Delta P_k &= P_{t+k} - P_{t+k-1}, \quad k = 2, \dots, 16. \end{aligned} \quad (5)$$

With $\epsilon = 0.05P_{\text{rated}}$ and $P_{\text{rated}} = 30.1$ kW, the three labels are assigned as

$$y_k = \begin{cases} 0 \text{ (down)}, & \Delta P_k < -\epsilon, \\ 1 \text{ (stable)}, & |\Delta P_k| \leq \epsilon, \\ 2 \text{ (up)}, & \Delta P_k > \epsilon. \end{cases} \quad (6)$$

The median used by the slope-weighted regression loss is calculated independently along the time axis of each target sequence, with $\alpha_S = 1.0$. The auxiliary focal loss uses $\gamma = 2.0$. Each training clip is evaluated through two future-video views: the observed future sequence and its aligned PhyDiffNet-generated sequence. If $\mathcal{L}_{\text{PV}}^{\text{real}}$ and $\mathcal{L}_{\text{PV}}^{\text{gen}}$ denote the corresponding composite RaPVFormer losses, the training objective is

$$\mathcal{L}_{\text{train}} = 0.5\mathcal{L}_{\text{PV}}^{\text{real}} + 0.5\mathcal{L}_{\text{PV}}^{\text{gen}}. \quad (7)$$

Validation, testing, and deployment use only the generated future-video view.

D. Shared Data Split and Augmentation

All three model stages reuse one day-level split manifest, so clips from the same day cannot appear in different subsets. When a manifest is first generated, the released implementation uses random seed 42 and then persists the resulting date lists for every subsequent stage. For the 138-day dataset, this procedure assigns 110 days to training, 14 to validation, and 14 to testing.

The exact clip counts after quality control are those reported in the main text.

Spatial augmentation is enabled only for training clips. A sampled 90-degree rotation and optional horizontal and vertical flips are applied consistently to every time step and to all aligned RGB frames, solar-position masks, and generated frames. Validation and test clips are not augmented.

E. End-to-End Inference Protocol

The end-to-end inference path receives 16 historical RGB frames, 16 historical solar-position masks, 16 precomputed future solar-position masks, and 16 historical PV observations. PhyDNet first produces 16 coarse future RGB frames. The diffusion module then combines the last four historical RGB frames with the coarse sequence to generate 16 refined frames. Finally, RaPVFormer combines the historical observations, refined future frames, and corresponding masks to output 16 PV forecasts and the auxiliary ramp-class logits. Future observed RGB frames are excluded from validation, testing, and deployment at every stage.

III. METRICS FOR PREDICTED FULL-SKY VIDEO FRAMES EVALUATION

Detailed definitions of the frame-level and temporal metrics used to assess the quality and consistency of predicted full-sky video frames are provided in the following subsections.

A. Peak Signal-to-Noise Ratio (PSNR)

PSNR [1] is a traditional full-reference image quality metric that measures the pixel-wise similarity between the predicted frame $\hat{I}_t$ and the ground truth frame I_t . It is derived from the Mean Squared Error (MSE) between two images.

$$\text{PSNR}(I_t, \hat{I}_t) := 10 \log_{10} \left(\frac{L^2}{\text{MSE}(I_t, \hat{I}_t)} \right) \quad (8)$$

where L is the maximum pixel value and equals 255 for 8-bit images, and the MSE is defined as:

$$\text{MSE}(I_t, \hat{I}_t) := \frac{1}{HWC} \sum_{i=1}^{HWC} \left(I_t^{(i)} - \hat{I}_t^{(i)} \right)^2 \quad (9)$$

where H , W , C represent the height, width and number of channels of the images, respectively. $I_t^{(i)}$ and $\hat{I}_t^{(i)}$ are the pixel values at position i in the ground-truth and predicted images, respectively.

A higher PSNR value indicates better image prediction quality.

B. Structural Similarity Index (SSIM)

SSIM [2] measures the similarity by comparing their luminance, contrast, and structural information in local image patches. Unlike PSNR, which relies purely on pixel-wise error, SSIM is more consistent with human visual perception. The SSIM is computed as:

$$\text{SSIM}(I_t, \hat{I}_t) := \frac{(2\mu_{I_t}\mu_{\hat{I}_t} + C_1)(2\sigma_{I_t\hat{I}_t} + C_2)}{(\mu_{I_t}^2 + \mu_{\hat{I}_t}^2 + C_1)(\sigma_{I_t}^2 + \sigma_{\hat{I}_t}^2 + C_2)} \quad (10)$$

where μ_{I_t} , $\mu_{\hat{I}_t}$ are the mean pixel values; $\sigma_{I_t}^2$, $\sigma_{\hat{I}_t}^2$ are the variances; C_1 and C_2 are constants for stability, defined as $C_1 = (K_1 L)^2$ and $C_2 = (K_2 L)^2$; L is the maximum pixel value and equals 255 for 8-bit frames; $K_1 = 0.01$ and $K_2 = 0.03$ are default constants.

Higher SSIM values (closer to 1) indicate better structural similarity between the predicted and the ground-truth frames.

C. Learned Perceptual Image Patch Similarity (LPIPS)

LPIPS [3] measures the perceptual similarity by comparing deep feature representations extracted from a pretrained convolutional neural network such as VGG. Different from PSNR and SSIM, which operate in the pixel or structural domain, LPIPS reflects human perceptual judgments more accurately. LPIPS is defined as:

$$\text{LPIPS}(I_t, \hat{I}_t) := \sum_l \frac{1}{H_l W_l} \sum_{h,w=1}^{H_l, W_l} \left\| w_l \odot \left(f_l(I_t)_{hw} - f_l(\hat{I}_t)_{hw} \right) \right\|_2^2 \quad (11)$$

where $f_l(\cdot)$ denotes the feature map at the l -th layer of the network; H_l and W_l are the height and width of the feature map at layer l ; w_l is a learned weight tensor used to adjust the channel importance; $\odot$ denotes element-wise multiplication.

A lower LPIPS score indicates higher perceptual similarity. A perfect LPIPS score of 0 means the predicted and ground-truth frames are perceptually identical in the chosen feature space.

D. VGG-based Cosine Similarity (VGGCS)

VGGCS [4] is a semantic similarity metric that evaluates how close the predicted frame is to the ground-truth frame in the deep feature space of a pretrained VGG network. It captures high-level semantic content such as object shapes and scene layouts.

The cosine similarity is computed as:

$$\text{VGGCS}(I_t, \hat{I}_t) := \frac{f(I_t) \cdot f(\hat{I}_t)}{\|f(I_t)\|_2 \times \|f(\hat{I}_t)\|_2} \quad (12)$$

where $f(\cdot)$ is flattened feature vector extracted from a specific VGG layer.

A higher VGGCS value (closer to 1) indicates greater semantic similarity between the predicted and ground-truth frames.

E. Temporal Optical Flow Consistency (TOF)

TOF [5] evaluates the temporal motion consistency of predicted video frames by comparing optical flows between adjacent frames in the predicted sequence and the corresponding flows in the ground-truth sequence. It is defined as:

$$\text{TOF} := \frac{1}{T-1} \sum_{t=1}^{T-1} \left\| \varphi(I_t, I_{t+1}) - \varphi(\hat{I}_t, \hat{I}_{t+1}) \right\|_2 \quad (13)$$

where $\varphi(\cdot, \cdot)$ denotes the optical flow from frame I_t to I_{t+1} , estimated using a pretrained flow network, such as Recurrent All-Pairs Field Transforms (RAFT) [5].

A lower TOF value indicates more accurate and temporally consistent motion patterns.

F. Temporal Feature Change Distance (TFCD)

TFCD measures how similarly the visual features evolve over time in both sequences. It leverages high-level features extracted from a pretrained network, such as VGG, to assess how well the semantic content is preserved over time. It is calculated as:

$$\text{TFCD} := \frac{1}{T-1} \sum_{t=1}^{T-1} \left\| \left(f(I_{t+1}) - f(I_t) \right) - \left(f(\hat{I}_{t+1}) - f(\hat{I}_t) \right) \right\|_2 \quad (14)$$

where $f(\cdot)$ is flattened feature vector extracted from a specific VGG layer.

A lower TFCD value indicates that the generated video exhibits more temporally consistent and realistic motion relative to the real reference.

IV. DATASET AVAILABILITY

The present evaluation focuses on the single-site SKIPP'D dataset, dynamically evolving periods that are most relevant to minute-scale PV ramps. This choice reflects the need for minute-resolution PV measurements together with unobstructed full-sky imagery that preserves detailed sun-cloud geometry; several existing public datasets do not simultaneously satisfy these requirements. Future work will extend the evaluation to additional sites and operating conditions as suitable datasets become available. The sky images and photovoltaic power generation dataset (SKIPP'D) [6] for short-term solar forecasting from 2017 to 2019 are listed in Table IV.

TABLE IV
DATA SOURCE OF DIFFERENT YEARS

Year	Data source
2017	https://purl.stanford.edu/sm043zf7254
2018	https://purl.stanford.edu/tb002mq9407
2019	https://purl.stanford.edu/jj716hx9049

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